Total No. of Question : [4] **Registration No. :** | | | | | | | | | |

Programme Name : Bachelor of Computer Science & Engineering
S.Y.B.Tech Backlog Exam Sem.III(A.Y.2024-25) May 2025
III SEMESTER (2023 BATCH)
231CSEPCCL202-Computer Network

Duration : [04:00 PM - 06:00 PM]
Date : 07 May, 2025
Day : Wednesday
Marks : 50

Instructions :

(Q1) **Attempt all questions** [20.0]

(1.1) Draw OSI model and Explain data link layer and network layer. [7.0]

(1.2) What is Hamming code? A bit word 1011001 is to be transmitted. Construct Hamming code for this data. [7.0]

(1.3) How to extract/find number of addresses, first address and last address in classless addressing? One of the addresses in a block is 110.23.120.14/20. Find number of addresses in a network, the first address and the last address. [6.0]

(Q2) **Attempt any two questions** [10.0]

(2.1) List options used by IP datagram. Explain any two multiple byte options. [5.0]

(2.2) Describe redirection message used by ICMP. [5.0]

(2.3) Explain Dijkstras algorithm to compute the shortest path with example. [5.0]

(Q3) **Attempt any two questions** [10.0]

(3.1) Explain following transport layer services: [5.0]
a. Error control
b. Congestion control

(3.2) Explain UDP advantages, disadvantages, and applications. [5.0]

(3.3) Write a short note on retransmission timer. [5.0]

(Q4) **All questions are compulsory.** [10.0]

(4.1) Define DNS. Explain in brief working of DNS. [5.0]

(4.2) Illustrate the working of FTP along with its types of connections. [5.0]